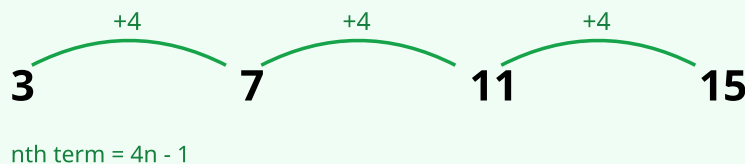


Fibonacci-Type Sequences

AQA · KS3 Year 7–8

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

- Multiplying instead of adding.
- Adding the wrong pair of terms.
- Missing term: term = (term before) + (two before).
- Arithmetic slips with larger numbers.

METHOD

- 1 Each term = sum of the two before it
- 2 Continue by adding the last two
- 3 Missing term: use term = prev + prev-prev
- 4 Work backwards by subtracting
- 5 Check the rule everywhere

WORKED EXAMPLE

Continue 1, 1, 2, 3, 5, 8, ...

Next = $5 + 8 = 13$, then $8 + 13 = 21$.

Backwards check: $13 - 8 = 5$

VISUAL AID

Each term is built from the two before it: 1, 1, 2, 3, 5, 8, 13, ... This is the famous Fibonacci pattern, and the ratio of neighbouring terms approaches the golden ratio (1.618).

1. (2 marks)

Write the next two terms: a) 1, 1, 2, 3, 5, 8, ... b) 2, 2, 4, 6, 10, ...

2. (2 marks)

Each term = sum of the two before. Write the next term: a) 3, 5, 8, ... b) 1, 4, 5, 9, ...

3. (3 marks)

Find the missing term: a) 1, __, 3, 5, ... b) 4, __, 11, 18, ... c) __, 5, 8, 13, ...

4. (3 marks)

Write the first five terms: a) start 1, 3 b) start 2, 5 c) start 4, 4

5. (3 marks)

a) First two terms 1 and 2 — write the first six terms. b) 3rd term 10, 4th term 16 — find the 1st and 2nd terms.

6. (3 marks)

Rabbit pairs: months 1 and 2 each have 1 pair; each month = sum of previous two. a) Pairs in months 3–6. b) First month with more than 10 pairs?

7. (4 marks)

3rd term 7, 4th term 11. a) Find 1st and 2nd terms. b) First six terms. c) 7th term.

8. (5 marks)

5th term 22, 6th term 35. a) Work backwards to first four terms. b) Whole sequence (1–6). c) Why does it grow more like a geometric sequence?